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I. Multiple Choice Questions

1. Consider two wires X and Y. The radius of wire X is 3 times the radius of Y. If they are stretched by the same load then the stress on Y is

(a) equal to that on X

(b) thrice that on X

(c) nine times that on X

(d) Half that on X

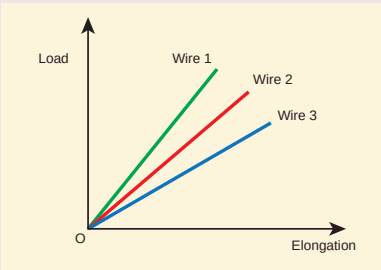
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2. If a wire is stretched to double of its original length, then the strain in the wire is

(a) 1 (b) 2

(c) 3 (d) 4

3. The load – elongation graph of three wires of the same material are shown in figure. Which of the following wire is the thickest?



(a) wire 1

(b) wire 2

(c) wire 3

(d) all of them have same thickness

4. For a given material, the rigidity modulus is $\left(\frac{1}{3}\right)^{rd}$ of Young's modulus. Its Poisson's ratio is

(a) 0 (b) 0.25

(c) 0.3 (d) 0.5

5. A small sphere of radius 2cm falls from rest in a viscous liquid. Heat is produced due to viscous force. The rate of production of heat when the sphere attains its terminal velocity is proportional to

(NEET model 2018)

(a) 2^2 (b) 2^3

(c) 2^4 (d) 2^5

6. Two wires are made of the same material and have the same volume. The area of cross sections of the first and the second wires are A and 2A respectively. If the length of the first wire is increased by Δl on applying a force F, how much force is needed to stretch the second wire by the same amount?

(NEET model 2018)

(a) 2 (b) 4

(c) 8 (d) 16

7. With an increase in temperature, the viscosity of liquid and gas, respectively will

(a) increase and increase

(b) increase and decrease

(c) decrease and increase

(d) decrease and decrease

8. The Young's modulus for a perfect rigid body is

(a) 0 (b) 1

(c) 0.5 (d) infinity